EXERCÍCIOS

(4) $F(x) = 2x^3 + 3x^2 - 36x$

$F(x)' = 6x^2 + 6x - 36$

$F(x)' = 6(x+3)(x-2)$

$x = -3$ e $x = 2$

(A) R: $x < -3 \to F(x)' > 0 \to$ CRESCENTE

R: $-3 < x < 2 \to F(x)' < 0 \to$ DECRESCENTE

R: $x > 2 \to F(x)' > 0$ CRESCENTE

CRESCENTE $(-\infty, -3)$ $(2, +\infty)$

(B) ~~[illegible]~~

$F(-3) = 2(-3)^3 + 3(-3)^2 - 36(-3) = 54$ $(-3, 54)$

$F(2) = 2(2)^3 + 3(2)^2 - 36(2) = -44$ $(2, -44)$

$x = -3$ CRESCENTE

$x = 2$ DECRESCENTE

(C)

$F(x)'' = 12x + 6$

PONTO $F(x)'' = 0 \to 12x + 6 = 0 \to x = -1/2$

$x < -1/2 \to F(x)''$

$x > -1/2 \to F(x)''$

$F(-1/2) = 37/2$

CONCAVA PARA BAIXO $(-\infty, -1/2)$

CONCAVA PARA CIMA $(-1/2, +\infty)$

PONTO DE INFLEXÃO $(-1/2, 37/2)$

(35) $f(x) = 2 + 2x^2 - x^4$

$f(x)' = 4x - 4x^3$

$f(x)' = 4x(1-x^2) = 4x(1-x)(1+x)$

$x = -1, 0, 1$

$(-\infty, -1) \to f(x)' > 0 \to$ CRESCENTE

$(-1, 0) \to f(x)' < 0$ DECRESCENTE

$(0, 1) \to f(x)' > 0$ CRESCENTE

$(1, +\infty) \to f(x)' < 0$ DECRESCENTE

(b) $f(x)'' = 4 - 12x^2$

$f(-1)'' = 4 - 12(1) = -8 < 0$

$f(0)'' = 4 > 0$

$f(1)'' = 4 - 12(1) = -8 < 0$

$f(-1) = 2 + 2(1) - 1 = 3$ $\quad (-1, 3)$ MAXIMO

$f(0) = 2$ $\quad (0, 2)$ MINIMO LOCAL

$f(1) = 3$ $\quad (1, 3)$ MAXIMO

(c) $f(x)'' = 4 - 12x^2 \to 0$

$\to 12x^2 = 4$ $\quad f(1/\sqrt{3}) = 2 + 2\left(\frac{1}{3}\right) - \left(\frac{1}{9}\right)$

$\to x^2 = 1/3$ $\quad = 2 + \frac{2}{3} - \frac{1}{9}$

$x = \pm 1/\sqrt{3}$

CONCAVA PARA BAIXO: $-8 < 0$

PARA CIMA: $4 > 0$

mmc = 9

$\to \frac{18}{9} + \frac{6}{9} - \frac{1}{9} = \frac{23}{9}$ $\quad \left(-\frac{1}{\sqrt{3}}, \frac{23}{9}\right)$ e $\left(\frac{1}{\sqrt{3}}, \frac{23}{9}\right)$